THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 4.47 cm²	Sector $= 25.67$ cm^2; triangle $= \frac{\sqrt{3}}{4}r^2 = 21.19$ cm^2; Segment $= 25.67 - 21.19 = 4.47$ cm^2
2	**(d)** 150 mm	Circumference $= \pi d = 66$ mm; diameters $= 4 \times 21 = 84$ mm; Total $= 150$ mm
3	**(a)** 1925 m²	Area $= \frac{180}{360} \times \frac{22}{7} \times 35^2$; Area $= 1925$ m^2
4	**(a)** 117 cm	Arc $= \frac{45}{360} \times 2\pi \times 42 = 33$ cm; Perimeter $= 33 + 84 = 117$ cm
5	**(c)** 30°	$\theta = \frac{22 \times 360}{2 \times \frac{22}{7} \times 42}$; $\theta = 30^\circ$
6	**(d)** 1400 cm²	Circle $= 3850$ cm^2; square $= 2450$ cm^2; Difference $= 1400$ cm^2
7	**(d)** 924 m²	Increase $= \frac{1}{4} \times \frac{22}{7}(35^2 - 7^2)$; Increase $= 924$ m^2
8	**(d)** 67.82 cm²	Sector $= 115.50$ cm^2; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2 \sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 47.68$ cm^2; Segment $= 115.50 - 47.68 = 67.82$ cm^2
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{1}{4} \times \frac{22}{7} \times 49 = 38.5$; R is the formula used.
10	**(c)** A is true but R is false.	Arc $= \frac{1}{6} \times 2 \times \frac{22}{7} \times 21 = 22$ cm: A true; arc length is $\frac{\theta}{360^\circ} \times 2\pi r$, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The 14 ribs divide the circle into 14 equal sectors; Area of the circle $= \frac{22}{7} \times 49^2 = 7546$ cm^2 (1); Area between ribs $= 7546 \div 14 = 539$ cm^2 (1).	1+1
12	Circumference $= 2 \times \frac{22}{7} \times 35 = 220$ cm; Distance 110 m $= 11000$ cm; revolutions $= \frac{11000}{220}$ (1); Revolutions $= 50$ (1).	1+1
13	Outer radius $= 38.5$ m; Area $= \pi(R^2 - r^2) = \frac{22}{7}(38.5^2 - 35^2)$ (1); Area of path $= 808.5$ m^2 (1).	1+1
14	The 18 ribs divide the circle into 18 equal sectors; Area of the circle $= \frac{22}{7} \times 63^2 = 12474$ cm^2 (1); Area between ribs $= 12474 \div 18 = 693$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of sector $= \frac{\theta}{360} \times \pi r^2$ (1); $\frac{50}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 770$ cm^2 (1).	1+1+1
16	The quadrant has radius s, so its area is $\frac{1}{4}\pi s^2$ (1); Quadrant $= 3118.5$ cm^2; square $= 3969$ cm^2 (1); Remaining $= 850.5$ cm^2 (1).	1+1+1
17	In 5 minutes the hand turns through $6^\circ \times 5 = 30^\circ$ (1); Area $= \frac{30}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 462$ cm^2 (1).	1+1+1
	OR: Area of sector $= \frac{\theta}{360} \times \pi r^2$ (1); $\frac{80}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 1232$ cm^2 (1).	1+1+1
18	The quadrant has radius s, so its area is $\frac{1}{4}\pi s^2$ (1); Quadrant $= 3850$ cm^2; square $= 4900$ cm^2 (1); Remaining $= 1050$ cm^2 (1).	1+1+1
	OR: In 10 minutes the hand turns through $6^\circ \times 10 = 60^\circ$ (1); Area $= \frac{60}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 924$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{60}{360} \times \frac{22}{7} \times 21^2 = \frac{1}{6} \times 1386 = 231$ m^2 (1). (ii) $\frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22$ m (1). (iii) $22 + 21 + 21 = 64$ m (1). (iv) $\frac{120}{360} \times 1386 = 462$ m^2 (1), which is double the old area, since the area is proportional to the angle (1).	1+1+1+2
20	Segment $=$ sector $-$ triangle (2); Sector $= 1848.00$ cm^2; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2 \sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 762.93$ cm^2 (2); Segment $= 1848.00 - 762.93 = 1085.07$ cm^2 (1).	2+2+1
	OR: Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{72}{360} \times \frac{22}{7} \times (21^2 - 14^2)$ (2); Area $= 154$ cm^2 (1).	2+2+1

Mistakes to avoid:

- Using πr^2 in the arc-length formula or $2\pi r$ in the sector-area formula.
- Forgetting to add the two radii when asked for the *perimeter* of a sector.

- Using diameter in place of radius in πr^2.
- Subtracting the wrong figure for a segment: minor segment $=$ sector $-$ triangle, not sector $+$ triangle.
- Mixing units (cm and m) or writing cm instead of cm^2 for areas.
- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 139 cm	Arc $= \frac{75}{360} \times 2\pi \times 42 = 55$ cm; Perimeter $= 55 + 84 = 139$ cm
2	**(c)** 60°	$\theta = \frac{44 \times 360}{2 \times \frac{22}{7} \times 42}$; $\theta = 60^\circ$
3	**(c)** 2016 cm²	Circle $= 5544$ cm^2; square $= 3528$ cm^2; Difference $= 2016$ cm^2
4	**(d)** 616 m²	Increase $= \frac{1}{4} \times \frac{22}{7}(35^2 - 21^2)$; Increase $= 616$ m^2
5	**(d)** 27.96 cm²	Sector $= 160.42$ cm^2; triangle $= \frac{\sqrt{3}}{4}r^2 = 132.45$ cm^2; Segment $= 160.42 - 132.45 = 27.96$ cm^2
6	**(d)** 86 mm	Circumference $= \pi d = 44$ mm; diameters $= 3 \times 14 = 42$ mm; Total $= 86$ mm
7	**(a)** 3773 m²	Area $= \frac{180}{360} \times \frac{22}{7} \times 49^2$; Area $= 3773$ m^2
8	**(b)** 183 cm	Arc $= \frac{135}{360} \times 2\pi \times 42 = 99$ cm; Perimeter $= 99 + 84 = 183$ cm
9	**(d)** A is false but R is true.	Area becomes $\pi(2r)^2 = 4\pi r^2$, four times: A false; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$\frac{22}{7} \times 7^2 = 154$ so A is true; R is true but not related to computing the area.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Circumference $= 2 \times \frac{22}{7} \times 49 = 308$ cm; Distance 154 m $= 15400$ cm; revolutions $= \frac{15400}{308}$ (1); Revolutions $= 50$ (1).	1+1
12	Outer radius $= 52$ m; Area $= \pi(R^2 - r^2) = \frac{22}{7}(52^2 - 45^2)$ (1); Area of path $= 2134$ m^2 (1).	1+1
13	The 20 ribs divide the circle into 20 equal sectors; Area of the circle $= \frac{22}{7} \times 70^2 = 15400$ cm^2 (1); Area between ribs $= 15400 \div 20 = 770$ cm^2 (1).	1+1
14	Circumference $= 2 \times \frac{22}{7} \times 56 = 352$ cm; Distance 176 m $= 17600$ cm; revolutions $= \frac{17600}{352}$ (1); Revolutions $= 50$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of sector $= \frac{\theta}{360} \times \pi r^2$ (1); $\frac{110}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 1694$ cm^2 (1).	1+1+1
16	The quadrant has radius s, so its area is $\frac{1}{4}\pi s^2$ (1); Quadrant $= 4658.5$ cm^2; square $= 5929$ cm^2 (1); Remaining $= 1270.5$ cm^2 (1).	1+1+1
17	In 20 minutes the hand turns through $6^\circ \times 20 = 120^\circ$ (1); Area $= \frac{120}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 1848$ cm^2 (1).	1+1+1
	OR: Area of sector $= \frac{\theta}{360} \times \pi r^2$ (1); $\frac{140}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 2156$ cm^2 (1).	1+1+1
18	The quadrant has radius s, so its area is $\frac{1}{4}\pi s^2$ (1); Quadrant $= 5544$ cm^2; square $= 7056$ cm^2 (1); Remaining $= 1512$ cm^2 (1).	1+1+1
	OR: In 25 minutes the hand turns through $6^\circ \times 25 = 150^\circ$ (1); Area $= \frac{150}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 2310$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{4} \times \frac{22}{7} \times 196 = 154$ cm^2 (1). (ii) $\frac{1}{2} \times 14 \times 14 = 98$ cm^2 (1). (iii) Minor segment $= 154 - 98 = 56$ cm^2 (1). (iv) Circle $= \frac{22}{7} \times 196 = 616$; major segment $= 616 - 56 = 560$ cm^2 (2).	1+1+1+2
20	Segment = sector − triangle (2); Sector $= 3285.33$ cm^2; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2 \sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 1356.32$ cm^2 (2); Segment $= 3285.33 - 1356.32 = 1929.01$ cm^2 (1).	2+2+1
	OR: Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{108}{360} \times \frac{22}{7} \times (21^2 - 14^2)$ (2); Area $= 231$ cm^2 (1).	2+2+1

Mistakes to avoid:

- Using πr^2 in the arc-length formula or $2\pi r$ in the sector-area formula.
- Forgetting to add the two radii when asked for the *perimeter* of a sector.
- Using diameter in place of radius in πr^2.
- Subtracting the wrong figure for a segment: minor segment = sector − triangle, not sector + triangle.
- Mixing units (cm and m) or writing cm instead of cm^2 for areas.

- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1232 m²	Increase $= \frac{1}{4} \times \frac{22}{7}(42^2 - 14^2)$; Increase $= 1232$ m²
2	**(a)** 753.52 cm²	Sector $= 1283.33$ cm²; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2 \sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 529.81$ cm²; Segment $= 1283.33 - 529.81 = 753.52$ cm
3	**(a)** 624 mm	Circumference $= \pi d = 176$ mm; diameters $= 8 \times 56 = 448$ mm; Total $= 624$ mm
4	**(d)** 4928 m²	Area $= \frac{180}{360} \times \frac{22}{7} \times 56^2$; Area $= 4928$ m²
5	**(b)** 161 cm	Arc $= \frac{105}{360} \times 2\pi \times 42 = 77$ cm; Perimeter $= 77 + 84 = 161$ cm
6	**(b)** 120°	$\theta = \frac{88 \times 360}{2 \times \frac{22}{7} \times 42}$; $\theta = 120^\circ$
7	**(d)** 3584 cm²	Circle $= 9856$ cm²; square $= 6272$ cm²; Difference $= 3584$ cm²
8	**(c)** 770 m²	Increase $= \frac{1}{4} \times \frac{22}{7}(42^2 - 28^2)$; Increase $= 770$ m²
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{22}{7}(196 - 49) = \frac{22}{7} \times 147 = 462$; R is the formula used.
10	**(c)** A is true but R is false.	In 30 minutes it turns through 180°, so area $= \frac{1}{2} \times 154 = 77$: A true; R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Outer radius $= 67$ m; Area $= \pi(R^2 - r^2) = \frac{22}{7}(67^2 - 60^2)$ (1); Area of path $= 2794$ m² (1).	1+1
12	The 24 ribs divide the circle into 24 equal sectors; Area of the circle $= \frac{22}{7} \times 84^2 = 22176$ cm² (1); Area between ribs $= 22176 \div 24 = 924$ cm² (1).	1+1
13	Circumference $= 2 \times \frac{22}{7} \times 42 = 264$ cm; Distance 198 m $= 19800$ cm; revolutions $= \frac{19800}{264}$ (1); Revolutions $= 75$ (1).	1+1
14	Outer radius $= 31.5$ m; Area $= \pi(R^2 - r^2) = \frac{22}{7}(31.5^2 - 28^2)$ (1); Area of path $= 654.5$ m² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area of sector $= \frac{\theta}{360} \times \pi r^2$ (1); $\frac{170}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 2618$ cm² (1).	1+1+1
16	The quadrant has radius s, so its area is $\frac{1}{4}\pi s^2$ (1); Quadrant $= 6506.5$ cm²; square $= 8281$ cm² (1); Remaining $= 1774.5$ cm² (1).	1+1+1
17	In 35 minutes the hand turns through $6^\circ \times 35 = 210^\circ$ (1); Area $= \frac{210}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 3234$ cm² (1).	1+1+1
	OR: Area of sector $= \frac{\theta}{360} \times \pi r^2$ (1); $\frac{200}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 3080$ cm² (1).	1+1+1
18	The quadrant has radius s, so its area is $\frac{1}{4}\pi s^2$ (1); Quadrant $= 7546$ cm²; square $= 9604$ cm² (1); Remaining $= 2058$ cm² (1).	1+1+1
	OR: In 40 minutes the hand turns through $6^\circ \times 40 = 240^\circ$ (1); Area $= \frac{240}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 3696$ cm² (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{22}{7} \times 28^2 = 2464$ m² (1). (ii) $\frac{22}{7} \times 35^2 = 3850$ m² (1). (iii) $3850 - 2464 = 1386$ m² (1). (iv) Cost $= 1386 \times 25 =$ ₹ 34,650 (1); inner boundary $= 2 \times \frac{22}{7} \times 28 = 176$ m (1).	1+1+1+2
20	Segment = sector − triangle (2); Sector $= 5133.33$ cm²; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2 \sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 2119.25$ cm² (2); Segment $= 5133.33 - 2119.25 = 3014.08$ cm² (1).	2+2+1
	OR: Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{180}{360} \times \frac{22}{7} \times (21^2 - 14^2)$ (2); Area $= 385$ cm² (1).	2+2+1

Mistakes to avoid:

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- Forgetting to add the two radii when asked for the *perimeter* of a sector.
- Using diameter in place of radius in πr^2.

- Subtracting the wrong figure for a segment: minor segment $=$ sector $-$ triangle, not sector $+$ triangle.
- Mixing units (cm and m) or writing cm instead of cm^2 for areas.
- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

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